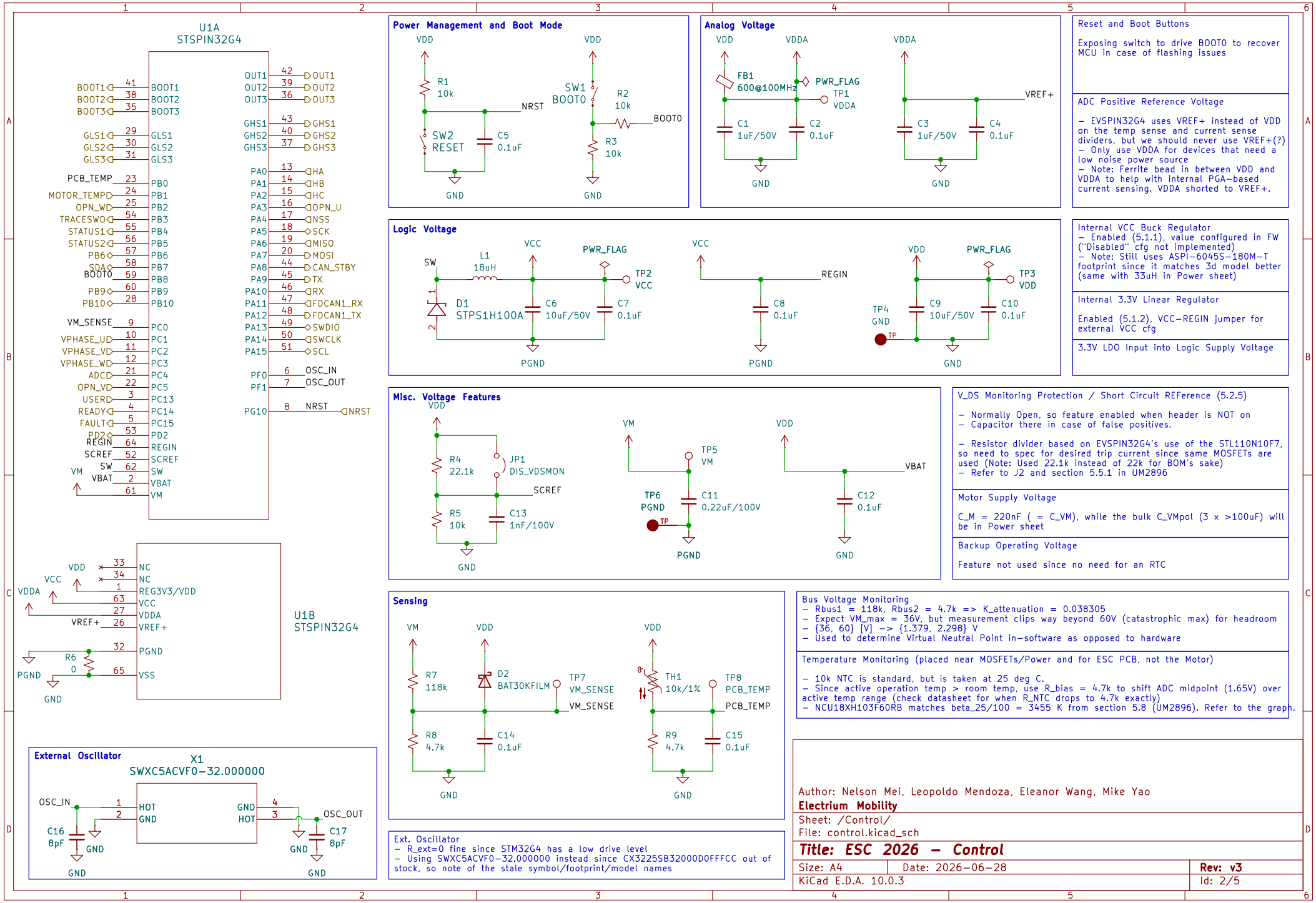

U1A
STSPIN32G4
BOOT1 41 BOOT1
BOOT2 38 BOOT2
BOOT3 35 BOOT3
GLS1 29 GLS1
GLS2 30 GLS2
GLS3 31 GLS3
PCB_TEMP 23 PB0
MOTOR_TEMP 24 PB1
OPN_W 25 PB2
TRACESWO 54 PB3
STATUS1 55 PB4
STATUS2 56 PB5
PB6 57 PB6
SDA 58 PB7
BOOT0 59 PB8
PB9 60 PB9
PB10 28 PB10
VM_SENSE 9 PC0
VPHASE_U 10 PC1
VPHASE_V 11 PC2
VPHASE_W 12 PC3
ADC 21 PC4
OPN_V 22 PC5
USER 3 PC13
READY 4 PC14
FAULT 5 PC15
PD2 53 PD2
REGIN 64 REGIN
SCREF 52 SCREF
SW 62 SW
VBAT 2 VBAT
VM 61 VM
OUT1 42 OUT1
OUT2 39 OUT2
OUT3 36 OUT3
GHS1 43 GHS1
GHS2 40 GHS2
GHS3 37 GHS3
PA0 13 HA
PA1 14 HB
PA2 15 HC
PA3 16 OPN_U
PA4 17 NSS
PA5 18 SCK
PA6 19 MISO
PA7 20 MOSI
PA8 44 CAN_STBY
PA9 45 TX
PA10 46 RX
PA11 47 FDCAN1_RX
PA12 48 FDCAN1_TX
PA13 49 SWDIO
PA14 50 SWCLK
PA15 51 SCL
PF0 6 OSC_IN
PF1 7 OSC_OUT
PG10 8 NRST
NRST
U1B
STSPIN32G4
VDD 33 NC
34 NC
VCC 1 REG3V3/VDD
VDDA 63 VCC
27 VDDA
VREF+ 26 VREF+
32 PGND
R6 0
PGND 65 VSS
GND
Power Management and Boot Mode
VDD
R1 10k
NRST
SW2 RESET
C5 0.1uF
GND
VDD
SW1 BOOT0
R2 10k
BOOT0
R3 10k
GND
Analog Voltage
VDD
VDDA
FB1 600@100MHz
PWR_FLAG
TP1 VDDA
C1 1uF/50V
C2 0.1uF
GND
VDDA
VREF+
C3 1uF/50V
C4 0.1uF
GND
Logic Voltage
SW
L1 18uH
VCC
PWR_FLAG
TP2 VCC
D1 STPS1H100A
C6 10uF/50V
C7 0.1uF
PGND
VCC
REGIN
C8 0.1uF
PGND
VDD
PWR_FLAG
TP3 VDD
TP4 GND
C9 10uF/50V
C10 0.1uF
GND
Misc. Voltage Features
VDD
R4 22.1k
JP1 DIS_VDSMON
SCREF
R5 10k
C13 1nF/100V
GND
VM
TP5 VM
TP6 PGND
C11 0.22uF/100V
PGND
VDD
VBAT
C12 0.1uF
GND
Sensing
VM
VDD
R7 118k
D2 BAT30KFILM
TP7 VM_SENSE
VM_SENSE
R8 4.7k
C14 0.1uF
GND
VDD
TH1 10k/1%
TP8 PCB_TEMP
PCB_TEMP
R9 4.7k
C15 0.1uF
GND
External Oscillator
X1
SWXC5ACVF0-32.000000
OSC_IN
1 HOT
2 GND
GND 4
HOT 3
OSC_OUT
C16 8pF
GND
GND
C17 8pF
GND
GND
Reset and Boot Buttons
Exposing switch to drive BOOT0 to recover MCU in case of flashing issues
ADC Positive Reference Voltage
– EVSPIN32G4 uses VREF+ instead of VDD on the temp sense and current sense dividers, but we should never use VREF+(?)
– Only use VDDA for devices that need a low noise power source
– Note: Ferrite bead in between VDD and VDDA to help with internal PGA–based current sensing. VDDA shorted to VREF+.
Internal VCC Buck Regulator
– Enabled (5.1.1), value configured in FW ("Disabled" cfg not implemented)
– Note: Still uses ASPI–6045S–180M–T footprint since it matches 3d model better (same with 33uH in Power sheet)
Internal 3.3V Linear Regulator
Enabled (5.1.2), VCC–REGIN jumper for external VCC cfg
3.3V LDO Input into Logic Supply Voltage
V_DS Monitoring Protection / Short Circuit REFerence (5.2.5)
– Normally Open, so feature enabled when header is NOT on
– Capacitor there in case of false positives.
– Resistor divider based on EVSPIN32G4's use of the STL110N10F7, so need to spec for desired trip current since same MOSFETs are used (Note: Used 22.1k instead of 22k for BOM's sake)
– Refer to J2 and section 5.5.1 in UM2896
Motor Supply Voltage
C_M = 220nF (= C_VM), while the bulk C_VMpol (3 x >100uF) will be in Power sheet
Backup Operating Voltage
Feature not used since no need for an RTC
Bus Voltage Monitoring
– Rbus1 = 118k, Rbus2 = 4.7k => K_attenuation = 0.038305
– Expect VM_max = 36V, but measurement clips way beyond 60V (catastrophic max) for headroom
– {36, 60} [V] –> {1.379, 2.298} V
– Used to determine Virtual Neutral Point in–software as opposed to hardware
Temperature Monitoring (placed near MOSFETs/Power and for ESC PCB, not the Motor)
– 10k NTC is standard, but is taken at 25 deg C.
– Since active operation temp > room temp, use R_bias = 4.7k to shift ADC midpoint (1.65V) over active temp range (check datasheet for when R_NTC drops to 4.7k exactly)
– NCU18XH103F60RB matches beta_25/100 = 3455 K from section 5.8 (UM2896). Refer to the graph.
Ext. Oscillator
– R_ext=0 fine since STM32G4 has a low drive level
– Using SWXC5ACVF0–32.000000 instead since CX3225SB32000D0FFFCC out of stock, so note of the stale symbol/footprint/model names
Author: Nelson Mei, Leopoldo Mendoza, Eleanor Wang, Mike Yao
Electrium Mobility
Sheet: /Control/
File: control.kicad_sch
Title: ESC 2026 – Control
Size: A4
Date: 2026–06–28
Rev: v3
KiCad E.D.A. 10.0.3
Id: 2/5